

Development of the Aortic Arches

Last updated February 28, 2026  10 min read  ScholarRx

Adapted by: [Alexis Gillett](#)

Editors: [Russell Johnson](#), [Jordan Finn](#)

Contributor: [Nancy Halliday](#)

Reviewers: [Diptiman Bose](#), [Helen Ryals](#), [Carrie Elzie](#)

 Adapted from: [Development of the Aortic Arches](#)

Listen to this Brick

0:00 / 17:17 **1x**

Learning Objectives (3)

Versions

After completing this brick, you will be able to:

- 1 **Analyze** the sequential development and remodeling of the six embryonic aortic arches to trace how each contributes to specific adult arterial derivatives of the head, neck, and thorax.
- 2 **Apply** embryologic principles to explain how asymmetric regression of the aortic arches establishes the differing courses of the right and left recurrent laryngeal nerves.
- 3 **Create** a concept map linking aortic arch development, vascular asymmetry, and related clinical conditions such as coarctation of the aorta and recurrent laryngeal nerve injury.

**CASE CONNECTION**

TB is a 4-year-old girl on postoperative day 1 following surgical repair of a coarctation of the aorta. She recovers well initially and is extubated within a few hours. The next day, however, her parents notice that her voice is hoarse and weaker than usual. When the problem persists for several days, her mother becomes concerned and asks: “Why is her voice like that? Did something happen during surgery?”

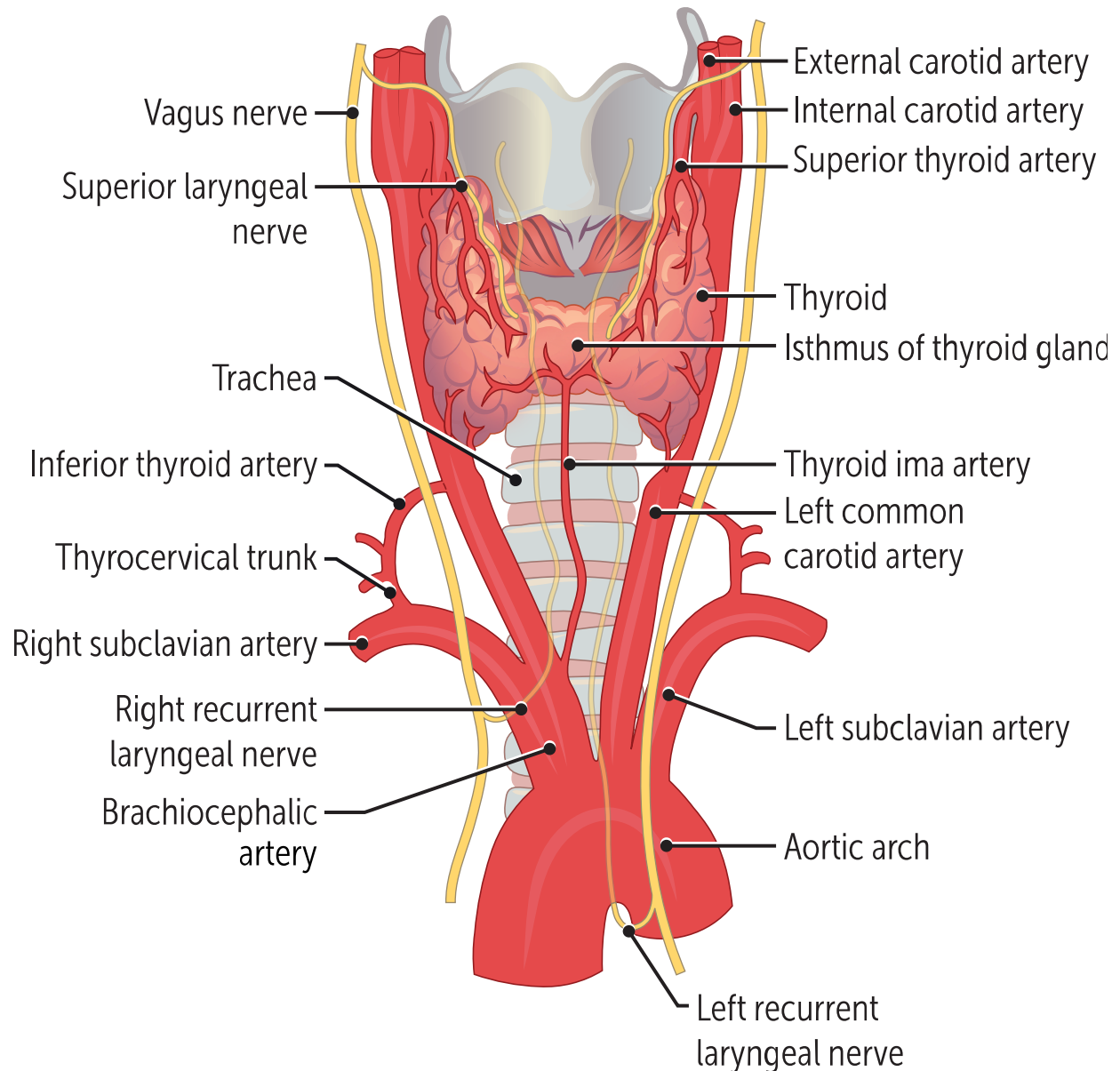
What is your leading diagnosis and how might it relate to her recent aortic repair? Keep your hypotheses in mind as you explore the development of the aortic arches and their nearby neural relationships—we’ll return to TB at the end of the Brick.

What Are the Aortic Arches? (LO#1)

The **embryonic aortic arches** are paired arteries that develop within the pharyngeal arches and connect the heart's outflow tract to the dorsal aortae. These transient vessels remodel extensively to form many of the **major arteries of the head, neck, and thorax** - including the common carotid arteries, aortic arch, subclavian arteries, pulmonary arteries, and ductus arteriosus.

Adult Aortic Anatomy

As shown in [Figure 1](#), the adult aorta gives rise to several major branches whose asymmetric arrangement reflects their embryologic origin. On the **right**, a brachiocephalic trunk divides into the right subclavian and right common carotid arteries. On the **left**, the common carotid and subclavian arteries arise separately from the aortic arch.



QUIZ

Tap image for quiz

Figure 1

This same region also reveals asymmetry in the **recurrent laryngeal nerves**.

- The right recurrent laryngeal nerve loops under the right subclavian artery.

- The left recurrent laryngeal nerve loops under the aortic arch, just behind the ligamentum arteriosum.

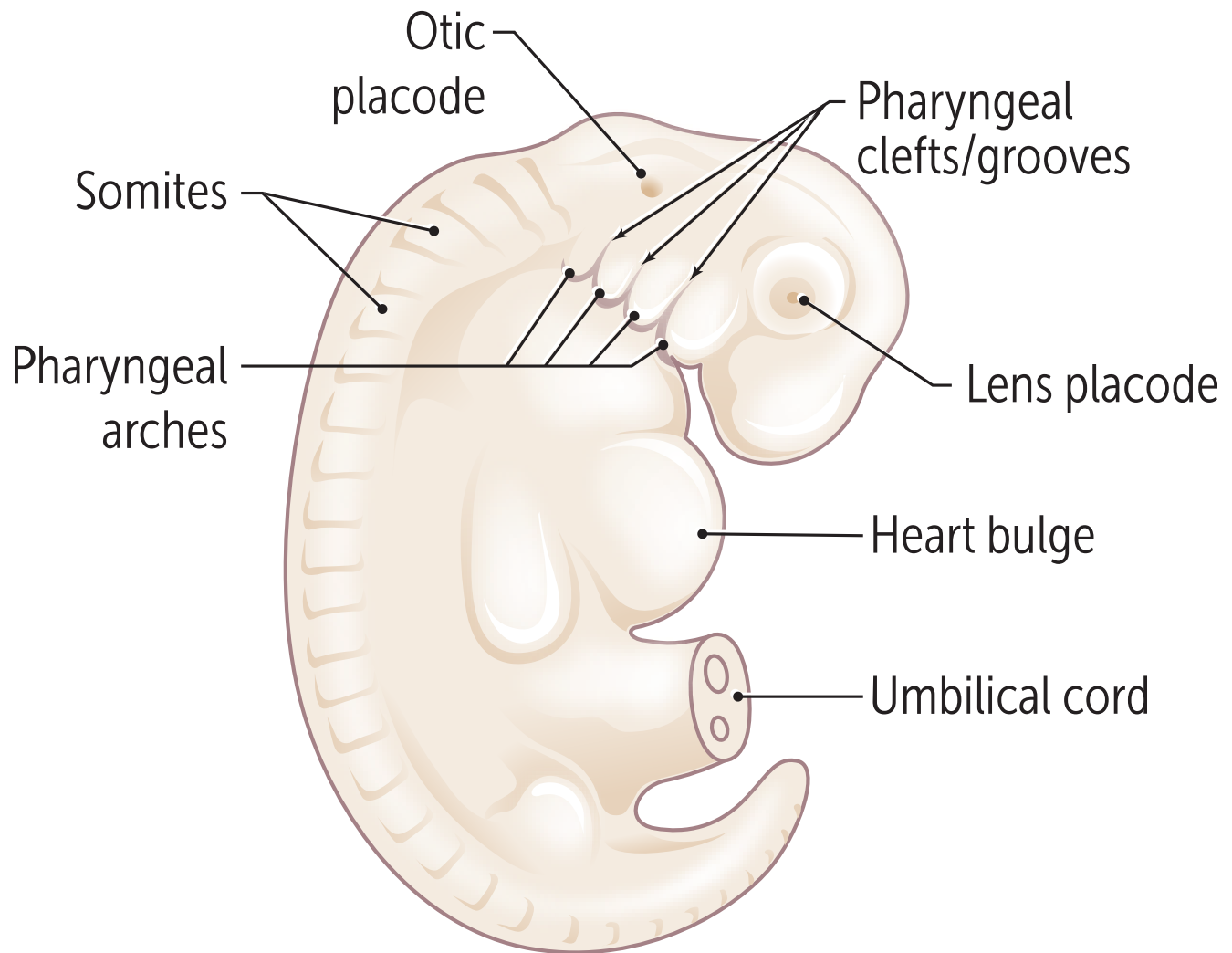
Understanding how these asymmetries arise requires tracing the development and remodeling of the embryonic aortic arches - a process that transforms an initially symmetric arterial pattern into the asymmetric configuration seen in the adult.

Pharyngeal Apparatus Development

During the fourth and fifth weeks of development, the embryo forms the pharyngeal apparatus (also called the branchial apparatus), which contributes extensively to the face and neck ([Figure 2](#)). The apparatus consists of six paired pharyngeal arches, arranged sequentially from cranial to caudal. Each arch contains a consistent set of components.

- **Cartilage bar** (largely derived from neural crest cells)
- **Muscle group**
- **Cranial nerve**
- **Artery** (called an aortic arch)

These embryonic aortic arches should not be confused with the single adult aortic arch, which is only one of their many derivatives.



QUIZ

 Tap image for quiz

Figure 2

A detailed discussion of the pharyngeal (branchial) arches can be found elsewhere. In this Brick, we focus specifically on how the paired arterial arches develop, remodel, and give rise to the asymmetric great vessels of the adult.

How Do the Aortic Arches Develop? (LO#1)

The **aortic arches** arise as paired arteries that branch from the **aortic sac**, the distal continuation of the truncus arteriosus at the outflow of the embryonic heart. Each arch passes dorsally through its corresponding **pharyngeal arch** to connect with the **paired dorsal aortae** (Figure 3).

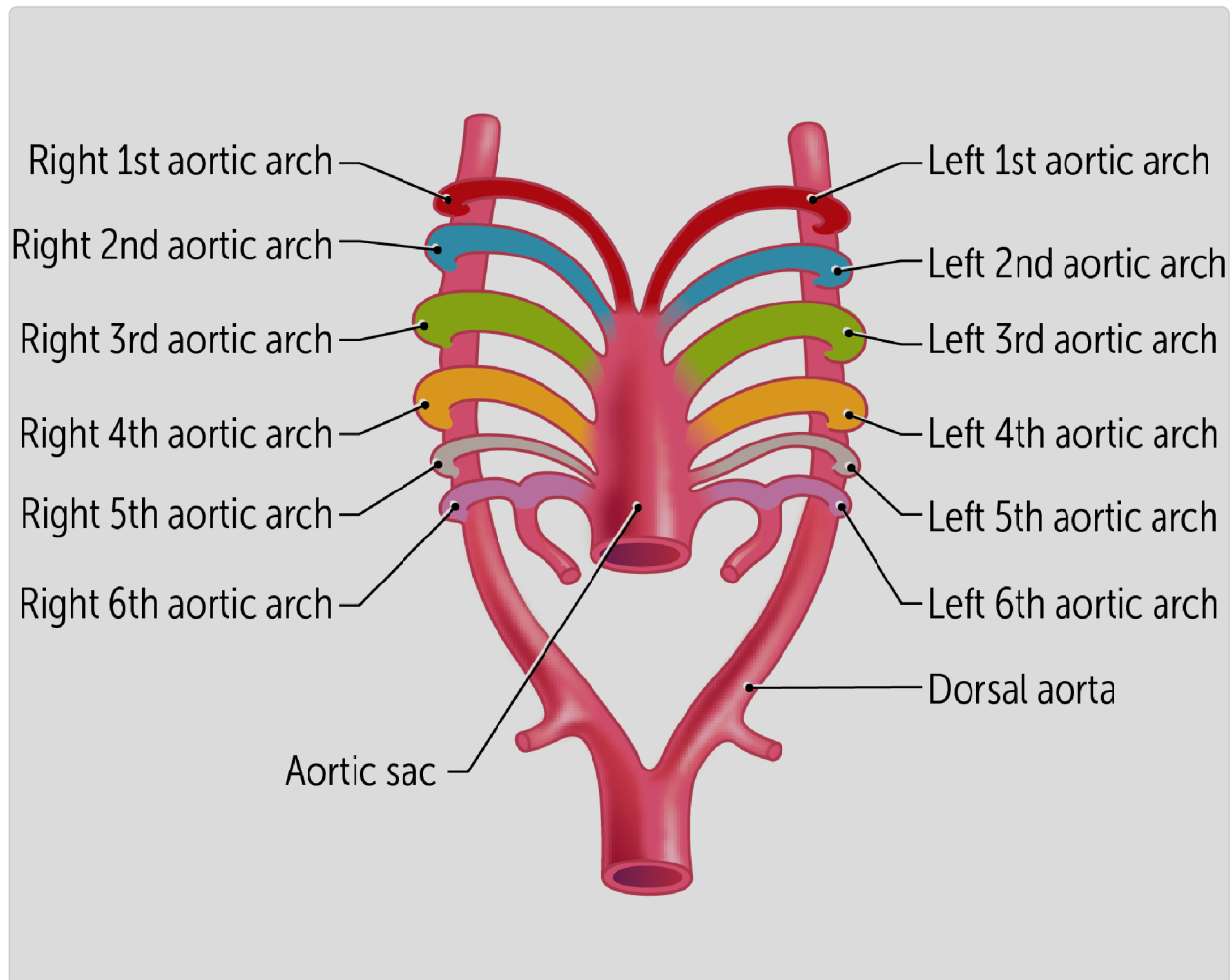


Figure 3

Six pairs of aortic arches form sequentially from head to tail, although the **fifth pair** soon regresses. These initially **symmetric vessels** remodel

extensively, establishing the framework for the **asymmetric great arteries** of the head, neck, and thorax seen in the adult.

the pharyngeal arches.

Major arteries of the head and neck arise from aortic arches found in

How Does Each Aortic Arch Develop into the Adult Pattern? (LO#1)

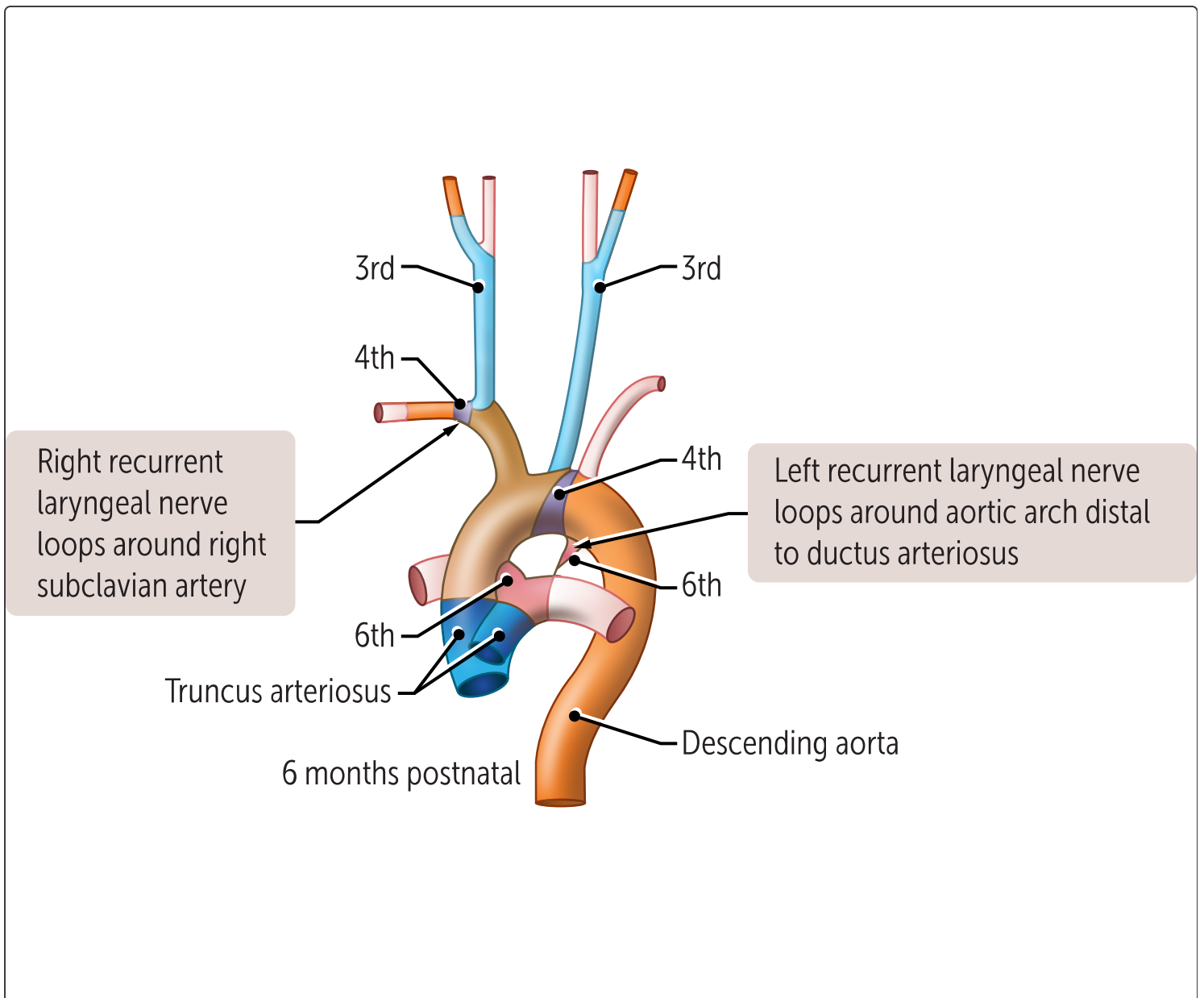
As development proceeds, the initially symmetric pairs of aortic arches undergo selective **regression, persistence, and remodeling** to produce the **asymmetric arterial pattern** of the adult great vessels. The fifth pair disappears early and contributes no adult structures.

Aortic Arch Derivatives

First and Second Aortic Arches. Mostly regress; however, small distal parts of the first pair persist and contribute to formation of maxillary arteries. The

second pair forms the transient stapelial arteries, which regress during development.

Third Aortic Arches. Form the common carotid arteries as well as the first part of the internal carotid arteries (Figure 4 & 5).



QUIZ

Tap image for quiz

Figure 4

arteries and the proximal internal carotid arteries.

The third aortic arch forms both the right and left common carotid

Fourth Aortic Arches. The right side becomes the proximal right subclavian artery, while the left side forms the arch of the aorta between the left common carotid (LCCA) and left subclavian arteries (LSA). Figure 4 & 5.

Sixth Aortic Arches. Right and left sides form the proximal right and left pulmonary arteries. On the left, the distal segment persists as the ductus arteriosus, connecting the pulmonary trunk to the aorta. Figure 4 & 5

ductus arteriosus.

The left sixth aortic arch forms the left pulmonary artery and the

Table 1 lists the pairs of aortic arches to provide a summary.

Table 1		
Aortic Arch	Right	Left
1st	Maxillary artery (part)	Maxillary artery (part)
2nd	Stapedial artery (transient; regress)	Stapedial artery (transient; regress)
3rd	Common carotid artery and proximal internal carotid artery	Common carotid artery and proximal internal carotid artery
4th	Proximal right subclavian artery	Aortic arch (between LCCA and LSA)
5th	Does not form	Does not form
6th	Proximal right pulmonary artery	Proximal left pulmonary artery and ductus arteriosus

Dorsal Aorta Derivatives

As the aortic arches remodel, the paired dorsal aortae also undergo major changes (**Figure 5**).

- **Caudally**, the two dorsal aorta fuse to form a single descending aorta.
- **Cranially**, the right dorsal aorta partially regresses, leaving the distal segment as part of the right subclavian artery, while the left dorsal aorta remains as the continuation of the descending aorta.

EMBRYOLOGIC DEVELOPMENT OF THE AORTIC ARCH

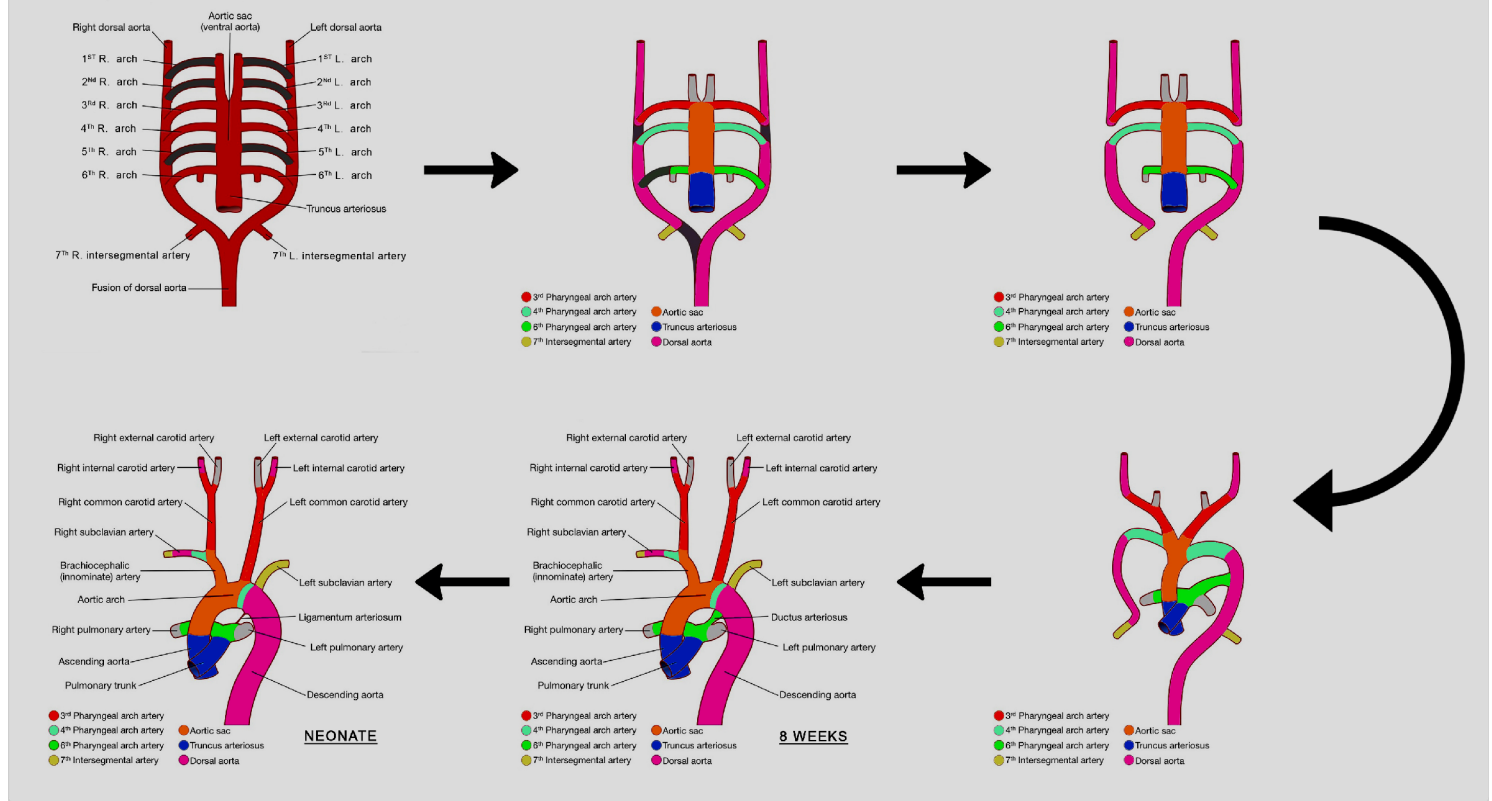


Figure 5

Other Modifications

At the same time, the aortic sac divides asymmetrically - forming the brachiocephalic trunk on the right and the ascending aorta on the left (see Figure 5).

The left subclavian artery does not originate from any aortic arches; it is instead a branch of the aorta.

Together, these changes finalize the asymmetric arterial configuration characteristic of the vessels in the adult thorax.

Coarctation of the aorta is constriction (narrowing) of the aortic lumen, usually located just distal to the origin of the left subclavian artery, near the site of the ductus arteriosus (ligamentum arteriosum). This characteristic location reflects the region's embryologic association with the left sixth aortic arch, where incorporation of ductal tissue and abnormal development (**partial involution**) **left dorsal aorta** lead to postnatal narrowing after ductal closure.

This results in **upper extremity hypertension with weak or delayed femoral pulses**, reflecting **lower blood pressure in the lower extremities compared with the upper extremities**. Severe cases present in neonates with heart failure and poor distal perfusion.

We can't discuss development of the aortic arches and their derivatives without also considering the nearby recurrent laryngeal nerves. Because these nerves develop in close proximity to the aortic arches, their final positions are shaped by the same vascular remodeling events. Let's examine how these nearby processes influence one another.

How Do the Recurrent Laryngeal Nerves Develop? (LO#2–3)

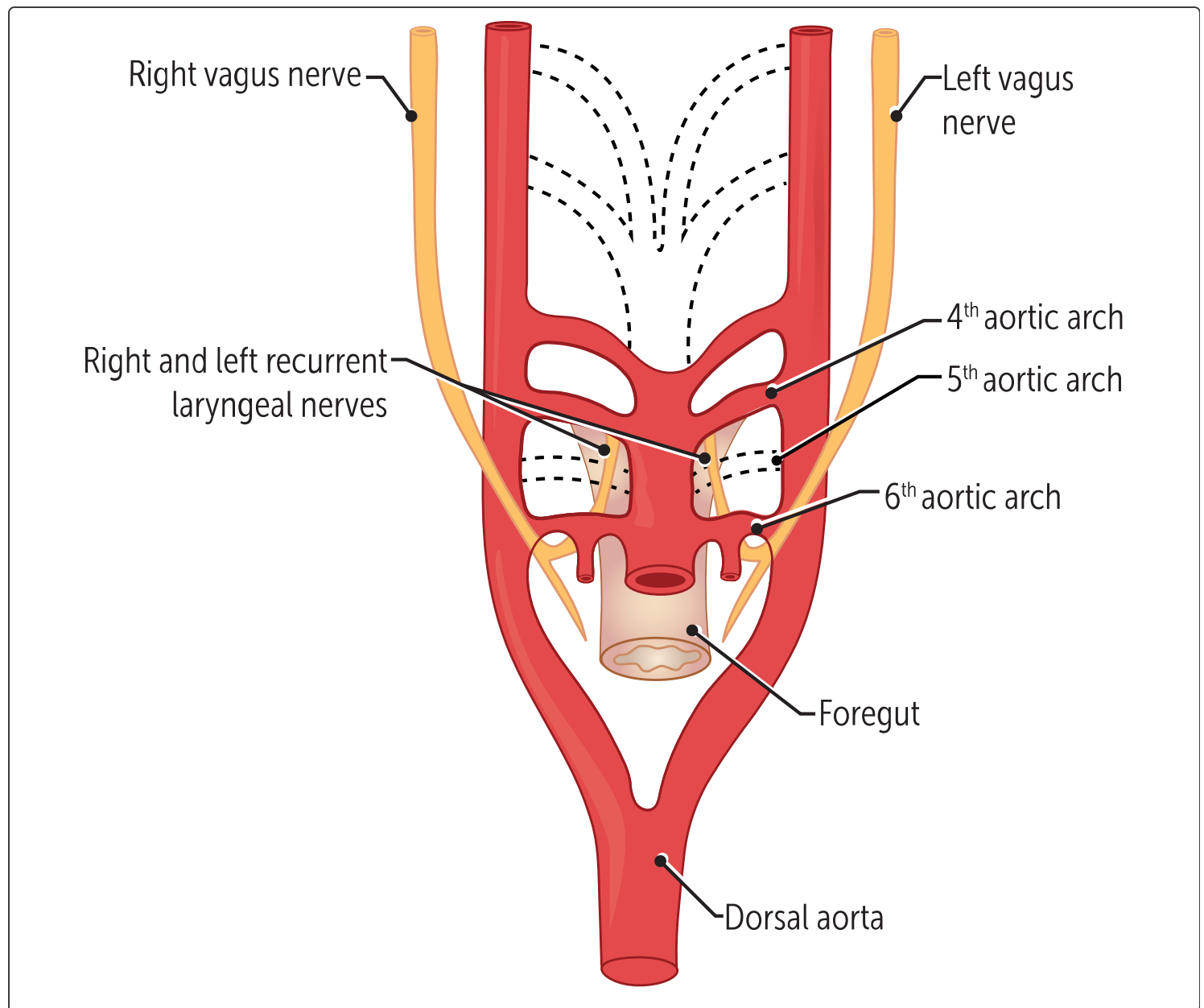
The **recurrent laryngeal nerves** are branches of the vagus nerve (**cranial nerve X**) that supply all the **intrinsic muscles of the larynx**, except the **cricothyroid**. In the adult, their **right and left courses differ markedly** because of changes in the surrounding **aortic arches** during development.

- The **right recurrent laryngeal nerve** loops under the right subclavian artery before ascending toward the larynx.
- The **left recurrent laryngeal nerve** loops under the arch of the aorta, just posterior to the ligamentum arteriosum.

This asymmetry reflects differences in the **sixth aortic arch remodeling**, which will be explored next.

Recurrent Laryngeal Nerve Asymmetry

During early development, the **right and left recurrent laryngeal nerves** both loop around the **sixth aortic arches** (Figure 6).



QUIZ



Figure 6

As the great vessels remodel, however, the **sixth arches develop asymmetrically**, and the nerves adjust their courses accordingly.

- On the **right**, the **distal portion of the sixth aortic arch regresses**. As the embryo elongates, the right recurrent laryngeal nerve moves upward and ultimately loops around the **right subclavian artery**, a derivative of the fourth aortic arch.
- On the **left**, the **sixth aortic arch persists** as the **ductus arteriosus**, maintaining its connection to the aortic arch. The left recurrent laryngeal nerve therefore remains looped under the **aortic arch**, adjacent to the **ligament arteriosum**.

courses because of the asymmetry of sixth aortic arch development.

The right and left recurrent laryngeal nerves follow asymmetric

 CLINICAL CORRELATION

Because of their close relationship to the **thyroid and parathyroid glands**, the **recurrent laryngeal nerves** are vulnerable during **neck surgeries**, especially thyroidectomy and parathyroidectomy. These nerves supply all **intrinsic muscles of the larynx** except the **cricothyroid**, so injury causes **hoarseness, a weak or breathy voice, and an increased risk of aspiration**. Surgeons therefore identify and preserve these nerves carefully before ligating nearby vessels.

The **left recurrent laryngeal nerve**, which loops beneath the **aortic arch** near the **ligamentum arteriosum**, is also at risk during **aortic or ductus arteriosus repair**. Understanding its embryologic pathway explains why procedures in this region - such as correction of **coarctation of the aorta** - can result in postoperative vocal changes.

 CASE CONNECTION[BACK TO INTRODUCTION](#) 

Thinking back to TB, what is your diagnosis? How did this occur?

The most likely diagnosis is **left recurrent laryngeal nerve (RLN) injury** resulting from her **aortic arch repair**.

Although postoperative **vocal cord edema** is possible, persistent **hoarseness** after several days points instead to RLN involvement. The left RLN is especially vulnerable during such procedures because it **loops beneath the aortic arch**, a relationship established by **persistence of the left sixth aortic arch** during development.

Fortunately, TB's nerve injury was **neuropraxic rather than permanent**, and her voice gradually returned to normal over the next six months. Her recovery

illustrates how embryologic relationships among the aortic arches and nearby nerves continue to have important clinical implications long after birth.

Summary

- The embryonic aortic arches are paired arteries within the pharyngeal arches that remodel to form the major arteries of the head, neck, and thorax.
- Although six pairs initially appear, the **fifth pair regresses**, and selective persistence of others produces the asymmetric pattern of the adult great vessels.
- The **left sixth aortic arch** forms the **ductus arteriosus**, while the corresponding right arch regresses - an event that explains the different paths of the **right and left recurrent laryngeal nerves**.
- Understanding these remodeling processes clarifies both normal anatomy and clinical conditions such as **coarctation of the aorta** and **recurrent laryngeal nerve injury**.

Review Questions

1. Which aortic arch is responsible for the formation of the ductus arteriosus?

- A. First
- B. Second

- C. Third
- D. Fifth
- E. Sixth

Explanation (requires correct answer)

2. The common carotid and proximal internal carotid arteries arise from which pair of aortic arches?

- A. First
- B. Second
- C. Third
- D. Fifth
- E. Sixth

Explanation (requires correct answer)

3. Which developmental event causes the **right recurrent laryngeal nerve** to loop under the **right subclavian artery** rather than the aortic arch?

- A. Degeneration of right sixth arch
- B. Asymmetry in aortic arch development
- C. Regression of the left sixth aortic arch
- D. Fusion of the dorsal aortae

Explanation (requires correct answer)



Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01

Analyze the sequential development and remodeling of the six embryonic aortic arches to trace how each contributes to specific adult arterial derivatives of the head, neck, and thorax.

02

Apply embryologic principles to explain how asymmetric regression of the aortic arches establishes the differing courses of the right and left recurrent laryngeal nerves.

03

Create a concept map linking aortic arch development, vascular asymmetry, and related clinical conditions such as coarctation of the aorta and recurrent laryngeal nerve injury.

Objective Confidence: 0/3